

Algorithm of the new approach on top of the Error Taxonomy

1. Run initial prompt on validation set and collect errors.
2. Build error taxonomy and group errors into categories.
3. Select important categories based on frequency and coverage.
4. Split categories into multiple overlapping clusters.
5. For each cluster (having a category set):
 - Generate few prompts by updating the initial prompt and keep the best one.
6. Select top prompts (each prompt comes from different clusters) based on their evaluation score and use them for inference.

Experiment Settings of Baseline (Error Taxonomy):

- Select top 7 categories or enough to reach 70% coverage of error counts.
- Generate 15 prompts.
- Select best 3 prompts for inference.

Experiment Settings of Clustered Approach:

- Use same selected categories by the baseline above.
- Divide the categories into 5 overlapping clusters.
- Generate 3 prompts per cluster (total = 15).
- Keep best 1 prompt per cluster.
- Select best 3 cluster prompts for inference.

Additional Details:

- Both methods generate 15 prompts (fair comparison).
- Each experiment is run 3 times and averaged the metrics.
- Also tested with broader setting: top 10 categories or up to 90% coverage.
- Model usage: Qwen3-14B for error category generation and Qwen3-4B for inference.

Results The clustered approach shows a small improvement based on the avg. of 3 runs. For the default setting, F1 improves from 31.9 to 33.28. For the broader setting (more errors used), F1 improves from 28.1 to 28.21.